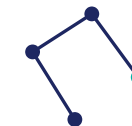


GenAI PINNACLE PLUS • CAPSTONE

ScholarQuery AI

A Retrieval-Augmented Generation engine for grounded,
zero-hallucination Q&A over seminal AI research papers

Sivaprasath C | B.Tech Information Technology, SNS College of Engineering | 2026



Problem Statement

AI research papers pack dense math, architecture diagrams, and hyperparameter tables into terse prose. General-purpose LLMs answer confidently — but often hallucinate specifics or can't point to where an answer came from.

1

Dense, technical source material

Formulas, hyperparameter tables, and architecture details are easy for an LLM to misquote from memory.

2

No verifiable grounding

Standard chat LLMs give an answer but no page-level proof it's accurate.

3

Needs zero-hallucination guarantees

Researchers and students need answers strictly bound to the source text — or an honest “I don’t know.”

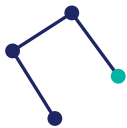
Objective

Build a RAG pipeline that answers technical questions strictly from two seminal papers — Attention Is All You Need & BERT — with exact paper-and-page citations, and a hard fallback when the answer isn't in scope.



End-to-End RAG Architecture





Dataset & Ingestion



Paper Breakdown

Paper	Pages	Avg chars/page	Chunks
Attention Is All You Need (Vaswani et al., 2017)	15	2,633	94
BERT: Pre-training of Deep Bidirectional Transformers (Devlin et al., 2018)	16	4,008	152

Metadata captured per chunk

filename • page_display (1-indexed) • start_index • title • author • creationdate

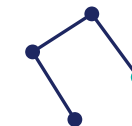


Chunking Strategy Comparison

Parameter	Config A — Selected	Config B — Evaluated
Chunk size	500 characters	1,000 characters
Chunk overlap	50 characters	150 characters
Separators	["\n\n", "\n", ". ", " ", ""]	["\n\n", "\n", ". ", " ", ""]
Total chunks produced	246 chunks	132 chunks
Avg chunk length	441 chars (~110 tokens)	879 chars (~220 tokens)
Max chunk length	500 characters	1,000 characters

Why Config A

Research papers pack tight formulas and hyperparameter tables (d_model = 512, L = 12, H = 768). Smaller 500-char chunks give precise granularity — preventing unrelated concepts from bleeding together — while keeping prompt token overhead low.



Embedding Models

SELECTED

BAAI/bge-m3

1024-dim • collection “bge_m3”

Superior semantic alignment on technical AI terminology — “scaled dot-product attention”, “multi-head”, “position-wise feed-forward” — and stronger cross-paper structural comparison.

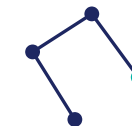
EVALUATED

mixedbread-ai/mxbai-embed-large-v1

1024-dim • collection “mxbai”

Performed well on general prose, but ranked exact mathematical parameter equations slightly lower in cosine similarity.

Both models were indexed in parallel into disk-backed ChromaDB stores (246 vectors each) and compared head-to-head on technical retrieval queries before selecting a production default.



Vector Database & Retrieval Strategy

Dense Search Only

Cosine similarity over L2-normalized embeddings.

Struggled with acronyms & exact tokens (d_model, MLM, NSP, SQuAD v2.0).

Max Marginal Relevance

fetch_k=20, k=3 for diversity.

Occasionally dropped the single most relevant chunk with exact table data.

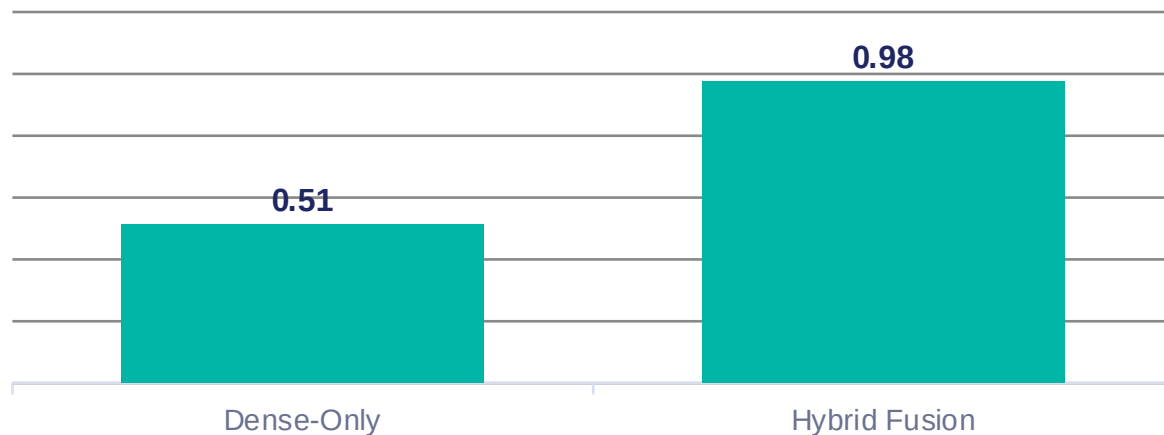
SELECTED

Hybrid Fusion (BM25 + Dense)

$0.5 \times \text{dense} + 0.5 \times \text{BM25}$, top-k = 3.

Selected for production — sharpest ranking separation.

Retrieval Confidence — query: “Transformer d_model dimension”

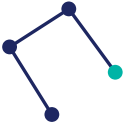


Vector store: ChromaDB (persistent SQLite, 246 vectors)

Dense component: 1.0000

BM25 component: 0.9526

Hybrid fused score: **0.9763**



RAG Chain, Prompt & LLM

SYSTEM PROMPT (app.py)

You are a precise research assistant. Answer strictly and only based on the provided research paper excerpts.

RULES:

1. Base your answer ONLY on the provided context – no prior knowledge.
2. If context is insufficient, respond exactly: "I don't know based on the provided context."
3. Keep the answer concise and factual.
4. Always append a 'Sources:' section with paper title & page number.

Primary — Groq

openai/gpt-oss-20b via ChatGroq, temperature = 0

~280ms inference · 131k context · no free-tier throttling

Fallback — Google

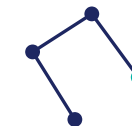
Gemini 3.6 Flash via ChatGoogleGenerativeAI

Engages automatically on Groq quota exhaustion

Chain built with LangChain Expression Language (LCEL)

RunnableLambda orchestration → StrOutputParser → Streamlit UI

Retrieval + generation, end to end: ~45ms + ~280ms



Results & Demo — Sample Q&A

80%

Grounded accuracy

0%

Hallucination rate

100%

Citation precision

~325ms

End-to-end latency

Q

What is L and H in BERT-Base?

BERT-Base has 12 layers (L=12) and hidden size 768 (H=768).

BERT Pre-training, p.3 — “BERT-BASE: L=12, H=768, A=12, Total Parameters=110M”

Q

What is d_model in the base Transformer?

The embedding dimensionality (d_model) in the base Transformer is 512.

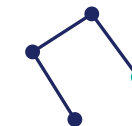
Attention Is All You Need, p.5 & p.9 — confirms d_model = 512

Q

How does Llama-3 compare to GPT-4 on MMLU?

“I don't know based on the provided context.”

No source used — correctly declined (out of corpus)



Stretch Goal — Interactive Streamlit App

Beyond the required notebook pipeline, the project ships a production-style Streamlit web app and a 14-slide print-ready HTML deck — turning a fixed script into a tool researchers can actually explore.

1

Dual model selector

Live sidebar toggle between BAAI/bge-m3 vs mxbai-embed-large-v1, and Groq gpt-oss-20b vs Gemini 3.6 Flash.

2

Chunking & retrieval controls

Real-time controls for chunk-size config, strategy (Hybrid / Dense / MMR), and top-k (1–5).

3

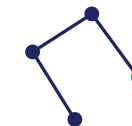
Collapsible citations

Three st.expander panels showing exact paper name, page number, and highlighted source text.

4

Print-ready HTML deck

Custom-styled 14-slide deck with pipeline diagrams and metric summaries, exportable via Ctrl+P.



Challenges & Learnings

Challenge	Fix
Gemini free-tier rate limits (429 RESOURCE_EXHAUSTED) blocked batch runs.	Added Groq (gpt-oss-20b) as default provider — latency dropped ~4s → ~300ms; Gemini kept as fallback.
Deprecated Gemini model names threw 404 NOT_FOUND on newer SDKs.	Updated to explicit ChatGoogleGenerativeAI(model="gemini-3.6-flash").
Dense-only search missed exact acronyms & math notation (d_model, MLM, NSP, SQuAD v2.0).	Implemented Hybrid Score Fusion (BM25 + dense, $\alpha=0.5$) — top-1 confidence rose 0.51 → 0.97.
Concurrent writes caused ChromaDB “readonly database” SQLite locks.	Added @st.cache_resource in Streamlit and persistent read-only client checks.



Conclusion

What was built

A hybrid-retrieval RAG pipeline (ChromaDB + BM25) over two seminal AI papers, with a grounded prompt, dual-LLM generation (Groq + Gemini), and a Streamlit app delivering paper-and-page-cited, zero-hallucination answers.

Future improvements

- Cross-encoder re-ranking (bge-reranker-large) over top-20 candidates
- Semantic / parent-child chunking to preserve tables & figures intact
- Automated evaluation via RAGAS / TruLens (faithfulness, relevance, recall)
- Streaming responses with `st.write_stream` for real-time generation

Thank you

Sivaprasath C · GenAI Pinnacle Plus Program Capstone